INFO 4220 – Networks II: Market Design
Spring 2022 – Problem Set 1

Due date: (Friday) February 11, 11.59PM EST

Late submissions will be penalized with 1 point for every hour late.

Instructions: For each question, show all your work or you will lose some or all of the credit for that question. ***Just showing the result without showing and explaining all steps you took to find it does not count.***

How to submit: Solutions must be submitted using Gradescope. For instructions on how to submit the assignment, go to this guide or watch this video.

Important: You must select the pages for each question in Gradescope. If you don't do this, you may be penalized with 3 points. If you don't know how to do this, read this guide.

Asking questions in Ed Discussions: You can use Ed Discussions to ask clarification questions about the problems. You cannot ask on Ed Discussions how to solve a particular problem, and specially, you cannot post on Ed Discussions (or anywhere else) your solution to the problem set (or a subset of it). Disclosing the solution of problems in Ed Discussions (or anywhere else) will be considered an academic integrity violation and reported accordingly. If you don't know how to solve a problem, don't ask on Ed Discussions, come to office hours! (Although, we will not solve the problem for you)

Who did you work with to solve this PS, if anyone (remember each of you must turn in their own solution, which has to be your own original work)? (0 pt)

Are you using late day wildcard for this PS? Remember that you have 2 for the entire semester, and they can only be used in 1-day increments. (0 pt)

2.5 Points for watching videos for L1 to L4 before February 4, 11.59PM EST

2.5 Points for your average score on the quizzes on each video for L2-L4

Problem 1 (10 points)

A year and a half into the COVID-19 pandemic, the supply and demand for N99 face masks has finally stabilized and they are being traded in a competitive market. The following table shows, for different prices, how many units sellers are willing to sell (supply), and how many units buyers are willing to buy (demand):

Price	Demand	Supply
12	26	2
15	20	5
18	14	8
21	8	11
24	2	14

a) Find the equation that represents the demand curve, and the equation that represents the supply curve (assume the demand curve and the supply curve are linear functions). (1 pt)

The demand curve is decreasing in price, thus it has the form: $Q_D = a - b * P$*. Finding a and b is easy by simply replacing a few values from the table to create a system of equations to find* $Q_D = 50 - 2P$

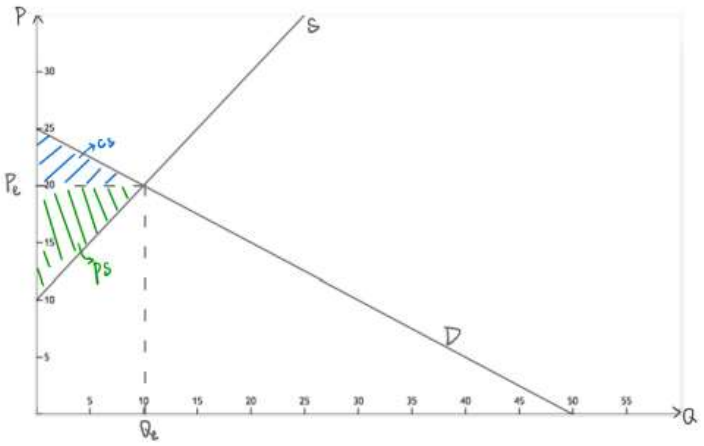
The supply curve is increasing in price, thus it has the form: $Q_S = a + b * P$*. Using value from the table it is to find that* $Q_S = -10 + P$

b) Draw a graph showing the demand curve and supply curve. (1 pts) What are the equilibrium price and quantities for N99 face masks, the consumer surplus, and the producer surplus? (2 pts) Please also denote equilibrium price, equilibrium quantity, consumer surplus and producer surplus in the graph. (1 pts)

To find the market equilibrium we set $Q_S = Q_D \Rightarrow -10 + P = 50 - 2P \Rightarrow P_e = 20, Q_e = 10$

The area corresponding to the consumer surplus is $CS = (25 - 20) * \frac{10}{2} = 25$

And the area corresponding to the producer surplus is $PS = (20 - 10) * \frac{10}{2} = 50$



c) Due to the Omicron wave, the demand for N99 face masks increases significantly. At every price, now the demand is twice as much as before. Draw the new demand curve and supply curve. Does the equilibrium price increases or decreases vs. the price calculated in (b) (1 pts)

If the demand is twice as before, the new demand curve is $Q_D' = 100 - 4P$

And the new market equilibrium: $Q_S = Q_D' \Rightarrow -10 + P = 100 - 4P \Rightarrow P_e' = 22, Q_e' = 12$

d) The government is concerned with the affordability of N99 face masks during the Omicron variant and decides to offer a rebate of $5 per mask (i.e. if a consumer pays $9 for a mask, they receive $5 back effectively making the price $4). Find the market equilibrium with the rebate (assuming the supply curve and the demand curve in part (c) (2 pts)

The new demand curve will be: $Q_D'' = 100 - 4 * (P - 5) = 120 - 4P$

And the new market equilibrium: $Q_S = Q_D'' = -10 + P = 120 - 4P \Rightarrow P_e'' = 26, Q_e'' = 16$

e) Explain what will happen to economic welfare: Compare the changes in consumer surplus and producer surplus with the total cost of the rebate program (with respect to the equilibrium in part (c)). Who is benefitting the most from the subsidy (buyers or sellers). (2 pts)

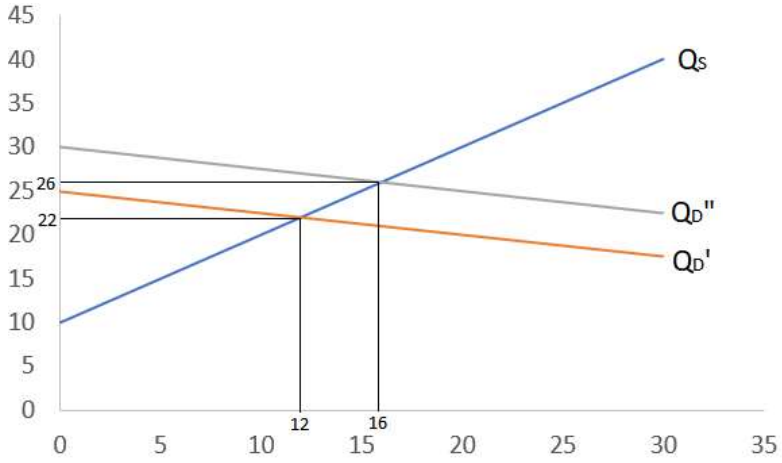
The following graph shows the equilibria from parts (c) and (d). The rebate shifts the demand curve up from Q_D' *to* Q_D''

Without the rebate: $PS = (22 - 10) * \frac{12}{2} = 72, CS = (25 - 22) * \frac{12}{2} = 18$

With the rebate: $PS = (26 - 10) * \frac{16}{2} = 128, CS = (30 - 26) * \frac{16}{2} = 32$

*The cost of the rebate program is: 16*5=80. Of this 80, 56 are captured by producers in the form of increased producer surplus, and 14 are captured by consumers as increased consumer*

surplus. There are 10 that are unaccounted and correspond to deadweight loss (i.e. loss of economic efficiency due to the intervention in the market).



Grading notes: Market equilibrium should include prices and quantities.

Problem 2 (12 points)

Suppose two players A and B play a game with the following payoff matrix:

		Player B	
		Left	Right
Player A	Up	$(a, 0)$	$(1-a, 1)$
	Down	$(1-a, 2)$	$(a, 1)$

Where a is an unknown constant between 0 and 1.

a) Do Players A and/or B have a dominant action in this game? Explain why or why not. (5 pts)

No one has a dominant action in this game. To see this, let's look at the best response function for each player:

For player B, the best response when Player A plays Up is to play Right. $R^B(Up) = Right$
For player B, the best response when Player A plays Down is to play Left. $R^B(Down) = Left$

For player A, the best response function will depend on both what player B does and the value of a*. When player B plays left,* $R^A(Left) =$

$$Up\ if\ a > \frac{1}{2}$$
$$Down\ if\ a < \frac{1}{2}$$

Note that they are indifferent with $a = \frac{1}{2}$*. Thus, it wouldn't be wrong if you define one of the relationship as not-strict.*

When player B plays right, $R^A(Right) =$

$$Up\ if\ a < \frac{1}{2}$$

$$Down\ if\ a > \frac{1}{2}$$

As the best response changes depending on what the other player does (for both A and B), then there are no dominant actions.

b) Identify any Nash equilibria which may exist. Note this may depend on the value of a. For any equilibria identified, prove they are a Nash equilibrium, and indicate whether there is an outcome of this game that is Pareto superior to said equilibrium. (7 pts)

A way of proving that an outcome $\hat{a}$ *is a Nash equilibrium is to show that for every player i, the action* $\widehat{a^i}$ *is the best response to the actions* $\widehat{a^{-i}}$ *of the other players. In this case, when* $a > 0.5$ *there will be no N.E. because at every possible outcomes one of the players will want to change its action. When* $a \leq 0.5$*:*

- *(Down, Left) is a Nash Equilibrium because* $R^A(Left) = Down$ *and* $R^B(Down) = Left$
- *(Up, Right) is a Nash Equilibrium because* $R^A(Right) = Up$ *and* $R^B(Up) = Right$

While there are two N.E. (Down, Left) is pareto superior to (Up, Right) because player B is better off in (Down, Left), and player A is indifferent between the two.

Problem 3 (12 points)

Two generals A and B are encamped on either side of a valley which is occupied by an enemy force. If both generals attack the enemy at the same time they will win. An attack by a single general, or an uncoordinated attack (i.e. if they don't attack at the same time), will fail.

Any general that attacks and fails will receive a utility of -1. A successful joint attack will yield a utility of 1 for each general. Not attacking yields a utility of 0 for the non-attacking general(s).

The year is 875, so communication via email or text is impossible. Furthermore, the Generals cannot use visual signals to coordinate the attack because their view is obstructed. Instead, it has been previously agreed that if an attack will be launched the next day, A will send a courier to B informing them of the time at which the attack will take place. Then the courier will return and confirm the message was delivered. However, there is some probability p (assume p is known to the generals) that the courier is a double agent and will give B a false time of attack. Assume that the courier may travel safely between the camps of A and B and that upon return the courier will always claim they have truthfully relayed the time of attack to B.

An attack message has been communicated and the generals must decide whether to attack or not. Find any Bayesian Nash equilibrium of this game and indicate how the decisions made by A and B will depend on p. Assume players use only pure strategies.

This is a game of incomplete information in which the payoff function if going to be determined by whether the messenger is a double agent or not.

With probability p the messenger is a double agent. When this happens, if the generals attack, they will lose because they will attack at the wrong time. The payoff function is:

		General B	
		Attack	Not Attack
General A	Attack	$(-1,-1)$	$(-1,0)$
	Not Attack	$(0,-1)$	$(0,0)$

With probability 1-p the messenger is not a double agent. When this happens, if the generals attack, they will win because they do it at the same time. The payoff function is:

		General B	
		Attack	Not Attack
General A	Attack	$(1,1)$	$(-1,0)$
	Not Attack	$(0,-1)$	$(0,0)$

To determine if a strategy is a Bayesian Nash Equilibrium, remember the definition:

A strategy profile s is a Bayesian-Nash Equilibrium is for each player i and each signal $t_i \in T_i$ and for each action $a_i \in A_i$: $E[u_i(s_i(t_i), s_{-i}(t_{-i}), (t_i, t_{-i})|t_i] \geq E[u_i(a_i, s_{-i}(t_{-i}), (t_i, t_{-i})|t_i]$

A strategy profile tells you what a player should do based on the signal they receive. In this case, the signal is the value of p. What the equation above is telling you is that a strategy profile in a B.N.E. no player has an incentive to deviate from the strategy profile (i.e. they won't get a greater expected payoff by using an action different from what the strategy profile tells them to do).

Note that (Not Attack, Attack) and (Attack, Not Attack) cannot be in an equilibrium because in both cases the General that attacks would have preferred not to attack (for any value of p). One general will always have an incentive to deviate.

The question then is, is (Attack, Attack) a Bayesian Nash Equilibrium or not?

The strategy profile here would be (the same for both Generals): If p is below p^* attack, if it is above not attack.

If Generals A and B attack, they will get an expected payoff of: $-1 * p + 1 * (1 - p) = 1 - 2p$

If Generals A (or B) deviates and don't attack while the other general sticks to the strategy of attacking (note we are trying to determine if any player has an incentive to deviate from the

strategy profile, assuming that the other players don't deviate), they get an expected payoff of: 0

Thus, for both generals to Attack if $p \leq p^*$ will be a B.N.E. (i.e. no one will have an incentive to deviate from the strategy profile stated above) as long as $1 - 2p \geq 0 \Rightarrow p \leq \frac{1}{2}$

Note that in this problem (Not Attack, No Attack) for any value of p is also a B.N.E. because none of the generals has an incentive to deviate.

If $p > 1/2$ the only possible N.E. is (Not Attack, Not Attack). In fact (Not Attack, Not Attack) is a B.N.E. for any p. However, (Attack, Attack) is the B.N.E. that gives the greater (or equal) payoff if $p \leq \frac{1}{2}$, so if the generals receive the attack signal and the probability of the messenger being a spy is less than 0.5, they should attack.

Problem 4 (11 points)

There are four musicians and four singers that are trying to form pairs. The preferences of musicians and singers are as follows:

P_{m1}	P_{m2}	P_{m3}	P_{m4}
s_1	s_1	s_4	s_2
s_4	s_2	s_3	s_1
s_3	s_4	s_2	s_4
s_2	s_3	s_1	s_3

P_{s1}	P_{s2}	P_{s3}	P_{s4}
m_2	m_1	m_1	m_4
m_4	m_4	m_2	m_1
m_1	m_3	m_3	m_2
m_3	m_2	m_4	m_3

a) Find a matching using the deferred acceptance algorithm with singers proposing. Show and explain each step of the algorithm (4 points)

Step 1: s_1 proposes to m_2, s_2 and s_3 propose to m_1, and s_4 proposes to m_4. Rejections: m_1 rejects s_2. At the end of the step, m_1 is temporarily matched to s_3, m_2 to s_1, m_3 is single, and m_4 to s_4.
Step 2: s_2 proposes to m_4. Rejections: m_4 (now holding offers from s_2 and s_4) rejects s_4. At the end of the step, m_1 is still temporarily matched to s_3, m_2 to s_1, m_3 is still single, and m_4 to s_2.
Step 3: s_4 proposes to m_1. Rejections: m_1 (now holding offers from s_3 and s_4) rejects s_3. At the end of the step, m_1 is now temporarily matched to s_4, m_2 to s_1, m_3 is still single, and m_4 to s_2.
Step 4: s_3 proposes to m_2. Rejections: m_2 (now holding offers from s_1 and s_3) rejects s_3. At the end of the step, m_1 is still temporarily matched to s_4, m_2 to s_1, m_3 is still single, and m_4 to s_2.
Step 5: s_3 proposes to m_3. There are no rejections as all musicians are matched to one (acceptable) singer. The temporary matches become final.
Matching: $\mu(m_1) = s_4$, $\mu(m_2) = s_1$, $\mu(m_3) = s_3$, $\mu(m_4) = s_2$

b) At each step of the algorithm, explain, for both musicians and singers, if their potential matchings are getting better or worse for them (3 points)

At step 1 of the algorithm, singers propose to their most preferred musician. Thus, those that are rejected and have to propose to a different musician in step 2 are going to get a matching that they like less than their first choice. If they get rejected again, their next possible matching will be one that they like less.

In the case of musicians, at each step their matching stays the same, or they get a matching that they like better. If a musician receives multiple offers on step 1, they keep the one that they like better. In steps 2 and above, they can only improve their matching, because if they get an offer from a singer they like better, they can reject their previous matching and keep the new offer.

c) Is the resulting matching stable? Is there any stable matching that would make musicians better off? (4 points)

The matching is going to be stable because the deferred acceptance algorithm in one-to-one matching problems always produces a stable result. Additionally, it produces the most preferred stable matching for the side that is proposing. In this case, singers where proposing, so the matching is the singers preferred matching. To know if the matching would be different if musicians propose we must do the algorithm again with musicians proposing. If we do that we get the same result, so there is no way of making musicians better off: $\mu(m_1) = s_4$, $\mu(m_2) = s_1$, $\mu(m_3) = s_3$, $\mu(m_4) = s_2$

Grading notes: In part c) is not enough to say that there is no way of making musicians better off. To get credit you have to either say that the best matching for musicians is if they propose and note that in this case the solution is the same, or run the algorithm with musicians proposing and find the matching (and you should find it is the same).